

ML & DL vs Buy & Hold

Stock Backtesting Report

A comparison of four trading strategies

Adobe | Microsoft | Oracle | Salesforce

Understanding how AI-based trading models performed against simply buying and holding the stock — explained in everyday language.

What Is This Report About?

Machine learning (ML) and deep learning (DL) models were trained on the historical stock exchange data of four major technology companies: **Adobe (ADBE)**, **Microsoft (MSFT)**, **Oracle (ORCL)**, and **Salesforce (CRM)**. Each model learned patterns from past price movements and made its own decisions on when to buy and when to sell.

To check whether these decisions were actually useful, each strategy was **backtested** — run against real historical price data to see how much money it would have made or lost. Each strategy's performance was then compared against the simplest possible approach: **Buy & Hold**, meaning buying the stock once at the start and simply holding it, without making any trading decisions at all.

This report explains, in everyday language, what each performance measure means, shows the results for all four companies side by side, and ends with a clear view on which company's results look most attractive for investment — along with the risks involved.

Quick Snapshot

Company	Model Used	Winner (Strategy or Buy & Hold)	CAGR	Max Drawdown
Adobe	Decision Tree	Strategy	22.13%	-43.40%
Microsoft	Logistic Regression	Buy & Hold	18.44%	-35.02%
Oracle	Random Forest	Buy & Hold	30.17%	-15.64%
Salesforce	Logistic Regression	Buy & Hold	5.64%	-56.84%

Understanding the Terms

Before looking at the numbers, here is what each measure actually means. These are standard stock-market and portfolio-analysis terms, explained simply.

1. Strategy Return / Final Value

This shows how much your money would have grown if you had followed the computer model's buy and sell signals. For example, if Rs. 100 became Rs. 454, your money grew about 4.5 times over the test period.

In stock-market terms: The cumulative return of the model-driven trading strategy, expressed as a growth multiple of the initial capital.

2. Buy & Hold Return

This is the simplest possible strategy: buy the stock once and never sell it until the end. It acts as the 'baseline' — if a fancy computer model cannot beat this simple approach, the model isn't adding real value.

In stock-market terms: The passive benchmark return, used to judge whether active/algorithmic trading outperformed a simple long-only position.

3. Sharpe Ratio

This tells you how much return you got for each unit of risk (ups and downs) you took on. A higher number is better — it means you were rewarded well for the risk, rather than just getting lucky.

In stock-market terms: Sharpe Ratio = (Return – Risk-free Return) ÷ Standard Deviation of Returns. It measures risk-adjusted performance using total volatility.

4. Sortino Ratio

This is similar to the Sharpe Ratio, but it only punishes the 'bad' kind of risk — the ups and downs that come from losses, not gains. A higher Sortino Ratio means the strategy handled downside moves well relative to the return it made.

In stock-market terms: Sortino Ratio = (Return – Risk-free Return) ÷ Downside Deviation. It measures risk-adjusted performance using only negative volatility.

5. Maximum Drawdown (Max DD)

This is the worst possible moment to have owned the investment — the biggest drop from a peak value to the lowest point that followed it, before recovering. A drawdown of -43% means that, at the worst point, the investment had lost 43% of its highest value.

In stock-market terms: Maximum Drawdown measures peak-to-trough decline over the backtesting period and is a key indicator of downside/tail risk.

6. CAGR (Compound Annual Growth Rate)

This converts the total return into a single 'per year' growth number, as if the investment grew by the same steady percentage every year. It makes it easy to compare investments that were tested over different lengths of time.

In stock-market terms: CAGR = (Ending Value ÷ Beginning Value)^(1 ÷ Number of Years) – 1. It represents the smoothed annualized growth rate of the investment.

Rule of thumb: Higher Strategy Return, higher CAGR, higher Sharpe Ratio, and higher Sortino Ratio are all good signs. A deeper (more negative) Maximum Drawdown is a bad sign, because it means bigger temporary losses along the way.

Company-by-Company Results

Each company below was tested with a different model, based on what performed best for that company's data. Figures assume an initial investment of Rs. 100 and ignore transaction costs, taxes, and slippage.

Adobe		Model tested: Decision Tree	
Measure	ML / DL Strategy	Buy & Hold	What it tells you
Model used	Decision Tree	—	Which method made the buy/sell decisions
Final value of Rs. 100	Rs. 454	Rs. 372	Money at the end of the test
Total return	354.0%	272.0%	Overall % gained
CAGR (yearly growth)	22.13%	—	Average growth per year
Sharpe Ratio	0.76	—	Return per unit of overall risk
Sortino Ratio	1.19	—	Return per unit of downside risk
Maximum Drawdown	-43.40%	—	Worst peak-to-bottom fall

Winner in this backtest: The Decision Tree strategy produced the higher final amount.

Adobe's own trading strategy (built using a Decision Tree model) was the only one of the four companies where the computer model actually beat simply buying and holding the stock. It turned Rs. 100 into Rs. 454, compared to Rs. 372 from Buy & Hold. This came with the largest drop of the four stocks along the way (-43.40%), so the ride was bumpy, but the destination was the best of any strategy tested.

Microsoft

Model tested: Logistic Regression

Measure	ML / DL Strategy	Buy & Hold	What it tells you
Model used	Logistic Regression	—	Which method made the buy/sell decisions
Final value of Rs. 100	Rs. 382	Rs. 433	Money at the end of the test
Total return	282.08%	332.94%	Overall % gained
CAGR (yearly growth)	18.44%	—	Average growth per year
Sharpe Ratio	0.80	—	Return per unit of overall risk
Sortino Ratio	1.27	—	Return per unit of downside risk
Maximum Drawdown	-35.02%	—	Worst peak-to-bottom fall

Winner in this backtest: Buy & Hold (simply holding the stock) produced the higher final amount.

Microsoft's stock on its own (Buy & Hold) grew faster than the Logistic Regression strategy built for it. Buy & Hold turned Rs. 100 into Rs. 433, while the strategy only reached Rs. 382. The strategy still made healthy, positive, risk-adjusted returns, and its drop from peak to bottom (-35.02%) was smaller than Adobe's, making it a somewhat calmer ride than Adobe.

Oracle

Model tested: Random Forest

Measure	ML / DL Strategy	Buy & Hold	What it tells you
Model used	Random Forest	—	Which method made the buy/sell decisions
Final value of Rs. 100	Rs. 133	Rs. 155	Money at the end of the test
Total return	32.90%	55.00%	Overall % gained
CAGR (yearly growth)	30.17%	—	Average growth per year
Sharpe Ratio	1.06	—	Return per unit of overall risk
Sortino Ratio	1.95	—	Return per unit of downside risk
Maximum Drawdown	-15.64%	—	Worst peak-to-bottom fall

Winner in this backtest: Buy & Hold (simply holding the stock) produced the higher final amount.

Oracle stands out for having the smoothest ride of all four companies. Its biggest drop from a peak was only -15.64%, by far the smallest of the group, while its yearly growth rate (CAGR of 30.17%) was the highest of

all four companies. Buy & Hold (Rs. 155) still edged out the Random Forest strategy (Rs. 133), but both numbers came with the best risk-adjusted scores (Sharpe 1.06, Sortino 1.95) in this entire comparison.

Salesforce

Model tested: Logistic Regression

Measure	ML / DL Strategy	Buy & Hold	What it tells you
Model used	Logistic Regression	—	Which method made the buy/sell decisions
Final value of Rs. 100	Rs. 124	Rs. 127	Money at the end of the test
Total return	24.49%	27.35%	Overall % gained
CAGR (yearly growth)	5.64%	—	Average growth per year
Sharpe Ratio	0.33	—	Return per unit of overall risk
Sortino Ratio	0.52	—	Return per unit of downside risk
Maximum Drawdown	-56.84%	—	Worst peak-to-bottom fall

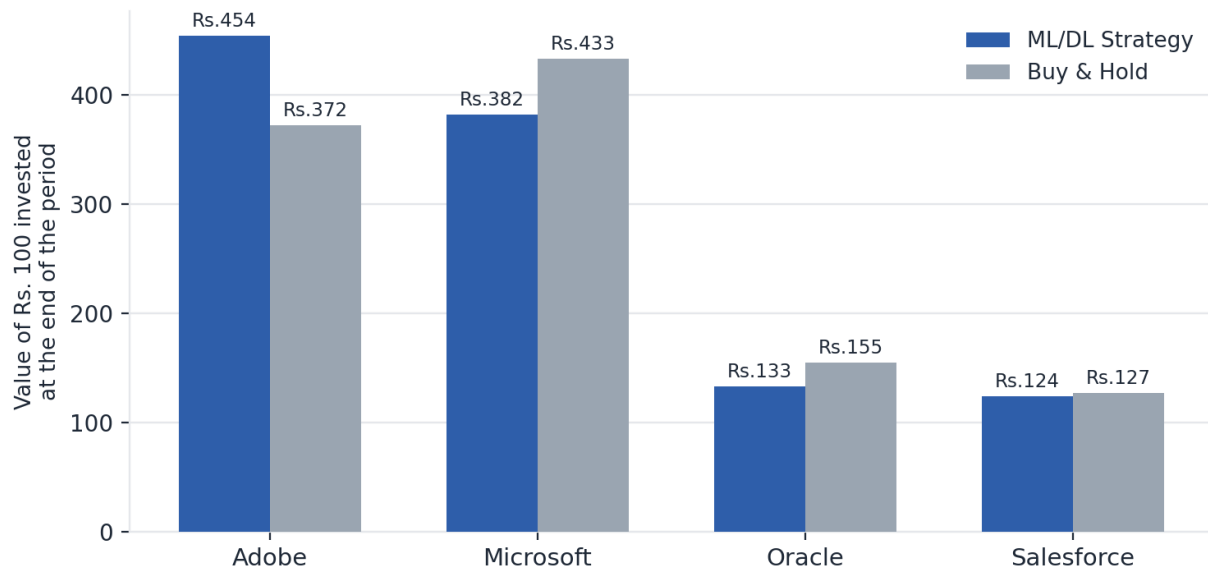
Winner in this backtest: Buy & Hold (simply holding the stock) produced the higher final amount.

Salesforce was the weakest performer in this comparison on almost every measure. Both the strategy and Buy & Hold barely grew the original investment (Rs. 124 and Rs. 127), the yearly growth rate was the lowest of the group (CAGR 5.64%), the risk-adjusted scores were the lowest (Sharpe 0.33, Sortino 0.52), and yet it still had the deepest drop of all four stocks (-56.84%). In short: low reward, high risk.

Comparing All Four Companies

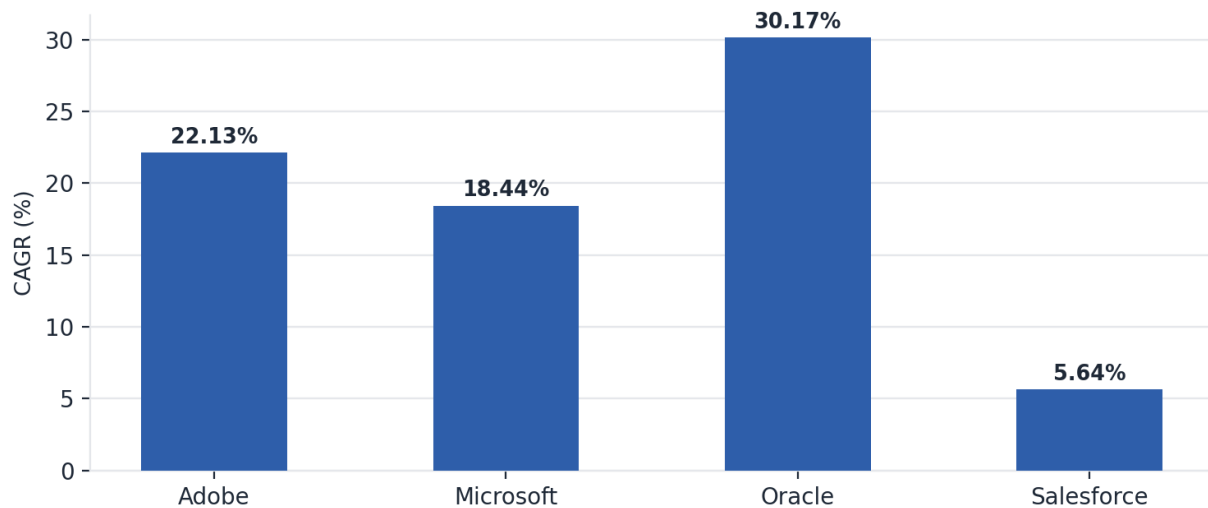
The charts below place all four companies side by side on the same measures, making it easy to see at a glance who performed best on returns, growth, risk-adjusted return, and downside risk.

Final Value of Rs. 100 Invested: ML Strategy vs Buy & Hold



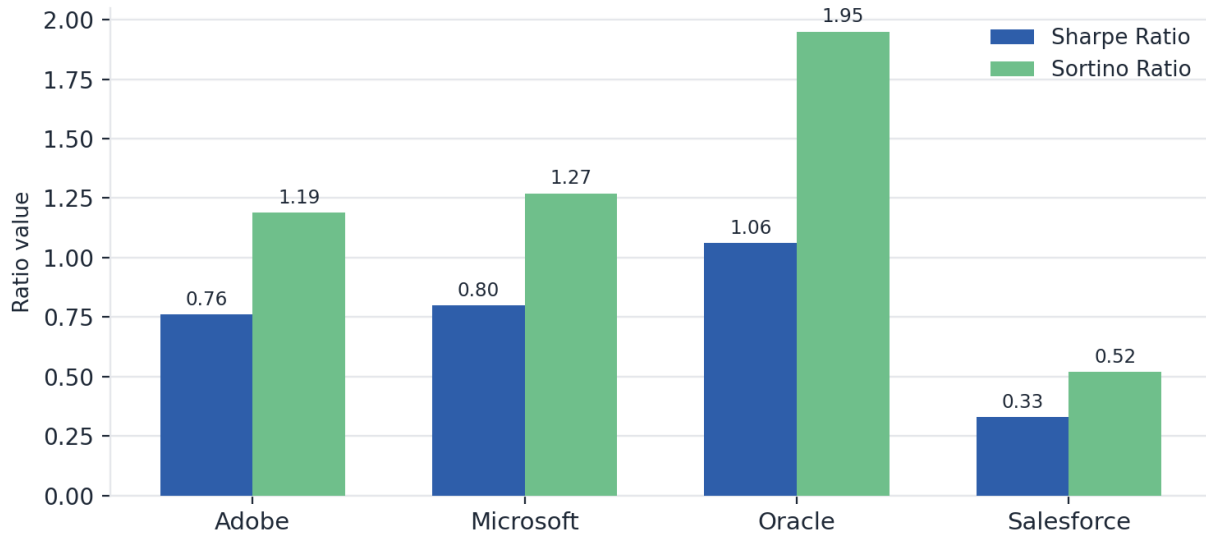
Adobe's strategy produced the highest overall multiple of the four (4.54x). Microsoft's Buy & Hold produced the single highest final value overall (4.33x), narrowly beating its own strategy. Oracle and Salesforce grew far less over the same style of test, though — as shown next — Oracle did so with much less risk.

Compound Annual Growth Rate (CAGR) of Each Strategy



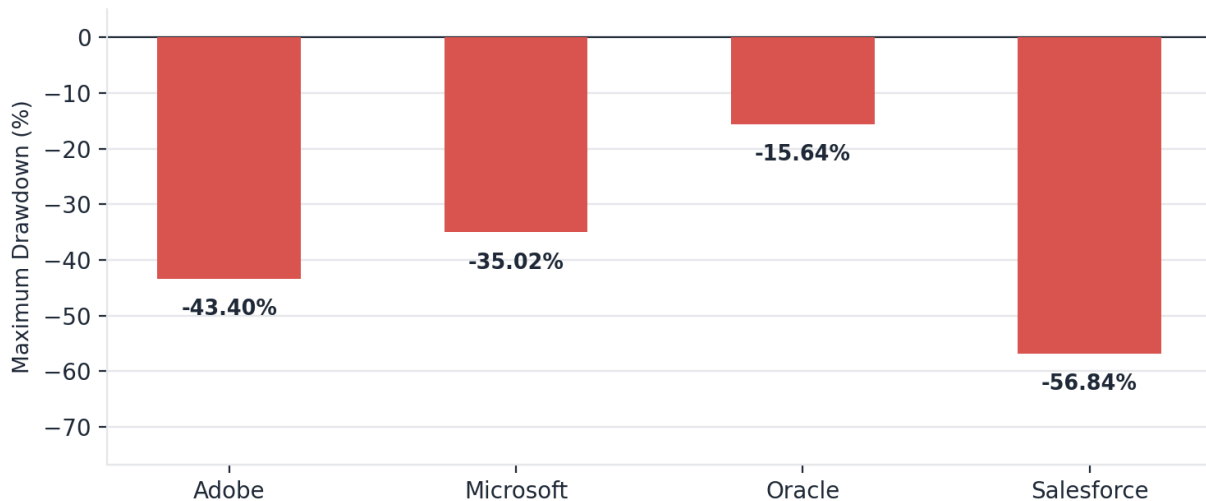
CAGR restates each result as a steady yearly growth rate, which is the fairest way to compare performance. On this measure, **Oracle's 30.17%** is actually the strongest annual growth rate of all four companies, ahead of Adobe (22.13%) and Microsoft (18.44%). Salesforce trails well behind at 5.64%.

Risk-Adjusted Return: Sharpe Ratio vs Sortino Ratio



Sharpe and Sortino ratios measure how much return was earned per unit of risk taken. **Oracle scores highest on both** (Sharpe 1.06, Sortino 1.95), meaning its strategy gave the best reward for the risk involved. Salesforce scores lowest on both, meaning it gave the weakest reward for its risk.

Maximum Drawdown: Biggest Drop From a Peak



Maximum Drawdown shows the worst temporary loss during the test. **Oracle had by far the smallest drawdown (-15.64%)**, meaning its value never fell too far from its peak. Salesforce had the deepest drawdown (-56.84%) — meaning an investor could have seen more than half their money temporarily wiped out at the worst point.

Side-by-Side Comparison Table

Measure	Adobe	Microsoft	Oracle	Salesforce
Model used	Decision Tree	Logistic Regression	Random Forest	Logistic Regression
Strategy final value (of Rs.100)	Rs. 454	Rs. 382	Rs. 133	Rs. 124
Buy & Hold final value (of Rs.100)	Rs. 372	Rs. 433	Rs. 155	Rs. 127
Strategy beat Buy & Hold?	Yes	No	No	No

CAGR	22.13%	18.44%	30.17%	5.64%
Sharpe Ratio	0.76	0.80	1.06	0.33
Sortino Ratio	1.19	1.27	1.95	0.52
Maximum Drawdown	-43.40%	-35.02%	-15.64%	-56.84%

Green highlights mark the best result in each row. Oracle leads on CAGR, Sharpe Ratio, Sortino Ratio, and (smallest) Maximum Drawdown — meaning it combined strong growth with the lowest risk of the four companies tested.

Final Verdict: Where Would Investing Make the Most Sense?

Best risk-adjusted pick: ORACLE

Looking purely at raw growth, Adobe's strategy produced the single biggest multiple (4.54x). But when you weigh growth **together with risk** — which is how professional investors actually compare opportunities — Oracle comes out on top. It delivered the highest annual growth rate (CAGR 30.17%), the best risk-adjusted returns (Sharpe 1.06, Sortino 1.95), and by far the smallest worst-case drop (-15.64%) of any company tested. That combination of strong growth and low downside risk makes Oracle the most attractive of the four for an investor who does not want to risk large temporary losses.

What if you want the highest possible growth and can accept more risk?

- **Adobe** is the strongest choice — its own trading strategy actually beat Buy & Hold, turning Rs.100 into Rs.454, the highest final value of any strategy in this comparison. The trade-off is a deep drawdown of -43.40% along the way.

Who to be cautious about:

- **Salesforce** is the weakest option in almost every measure — the lowest CAGR (5.64%), the lowest Sharpe and Sortino ratios, and the deepest drawdown (-56.84%) of all four companies. It combined the least reward with the most risk.
- **Microsoft** performed reasonably well overall, but its computer-driven strategy could not beat simply holding the stock — a reminder that a fancy model does not always add value over doing nothing.

Summary Ranking (Best to Weakest, Overall)

Rank	Company	Why
1	Oracle	Best risk-adjusted returns; highest CAGR; lowest drawdown
2	Adobe	Highest raw growth; strategy beat Buy & Hold; higher risk
3	Microsoft	Solid, steady performer; strategy lagged Buy & Hold
4	Salesforce	Weakest growth and risk-adjusted return; deepest drawdown

***Important note:** These results come from historical backtesting only — running the model against past price data. Backtested performance does not guarantee future results. Real trading also involves transaction costs, taxes, slippage, and changing market conditions that are not reflected here. This report is for educational and informational purposes only and should not be treated as financial or investment advice; consider speaking with a licensed financial advisor before making investment decisions.*